

TriageIQ

AI-assisted triage for IT support teams

Harnessing AI for Business

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AI impact research — a structured Who / Why / How view

WHO



Who: target customers and market

IT helpdesk teams in medium-to-large enterprises managing high volumes of repetitive support tickets — software, retail, education, financial services.

\$13.6B → \$106B

Helpdesk automation market (2025 → 2034)

70%

of CX leaders plan to integrate AI by 2027

WHY



Why: the operational pain AI can address

Manual triage takes 15-30 minutes per ticket and is inconsistent across agents. The result: missed escalations, SLA breaches, and bloated support costs. ML models classify priority consistently; generative AI drafts first responses at scale.

15-30 min

manual triage time per ticket

\$300K+/yr

labor cost at 500+ tickets/month

HOW



How: existing approaches and adoption gaps

Zendesk reports 85% AI-resolved tickets; Freshservice reports 66% deflection and 41% faster first response. Gartner expects 40% of enterprise apps to feature task-specific AI agents by 2026. Adoption barriers remain: limited internal AI expertise, insufficient training data, unclear ROI, and per-ticket pricing that doesn't scale.

40%

of enterprise apps to have AI agents by 2026 (Gartner)

4 barriers

expertise, data, ROI clarity, cost scalability

Market timing favors adoption — but most enterprises lack the expertise or data to deploy on their own.

THE PROBLEM

Manual ticket triage is slow, inconsistent, and expensive.



15–30 min

per ticket triaged



125–150 hrs

per month at 500+ tickets



\$300K+/yr

in manual triage labor cost

Beyond cost, manual triage is inconsistent.

Different agents read the same ticket differently. High-priority incidents get buried; low-priority requests escalate to senior staff. The result: SLA breaches, dissatisfied customers, and operational risk.

An AI-assisted layer over existing helpdesk platforms.

TriagelQ does not replace agents. It augments them — predicting priority, recommending routes, scoring SLA risk, and drafting first responses for human review. Designed to integrate with Jira Service Management, Freshservice, and ServiceNow.



01

Priority Detection

Classifies tickets as High, Needs Review, or Normal using TF-IDF text features. Optimized for recall on urgent issues — borderline cases flagged for human review.



02

Queue Routing

Predicts the right support group (Technical/ Product, Billing, etc.) with confidence thresholds: high-confidence routes auto-applied, lower-confidence sent to manual review.



03

SLA Risk Score

Transparent rule-based score combining predicted priority, complexity, customer tier, and history. Returns Low / Medium / High SLA risk with action recommendations.

Human-in-the-loop by design — every AI recommendation is reviewed before action.

How the three models are built and validated.

DATA

Eight datasets standardized into one pipeline

Multilingual support tickets (EN, DE) · IT ticket classification (200k records) · Customer support operational data · Jira issues + change history · Sample utterances

METHODOLOGY

Outer + inner holdout validation, model comparison, threshold tuning

Multiple candidate models trained per task. Inner holdout selects best model + threshold; outer holdout is held out until final reporting. Thresholds chosen against business objectives (false-negative rate ≤ 10% for priority).

01 Priority Detection

ALGORITHM

TF-IDF (word 1–3-grams, up to 70k features)

MODELS COMPARED

LogReg · SGD · Calibrated LinearSVC · XGBoost

KEY DESIGN DECISION

Two thresholds (Normal / High) create a "Needs Review" middle band — reduces false escalations while keeping recall high.

02 Queue Routing

ALGORITHM

TF-IDF + multi-class classifier

MODELS COMPARED

LogReg · SGD · ComplementNB · Calibrated LinearSVC

KEY DESIGN DECISION

Overlapping labels grouped into business-friendly routes. Three confidence buckets: auto-route · human review · manual triage.

03 SLA Risk Score

ALGORITHM

Rule-based scoring (ML compared, then rejected)

MODELS COMPARED

Logistic Regression · XGBoost — both ROC-AUC ≈ 0.50

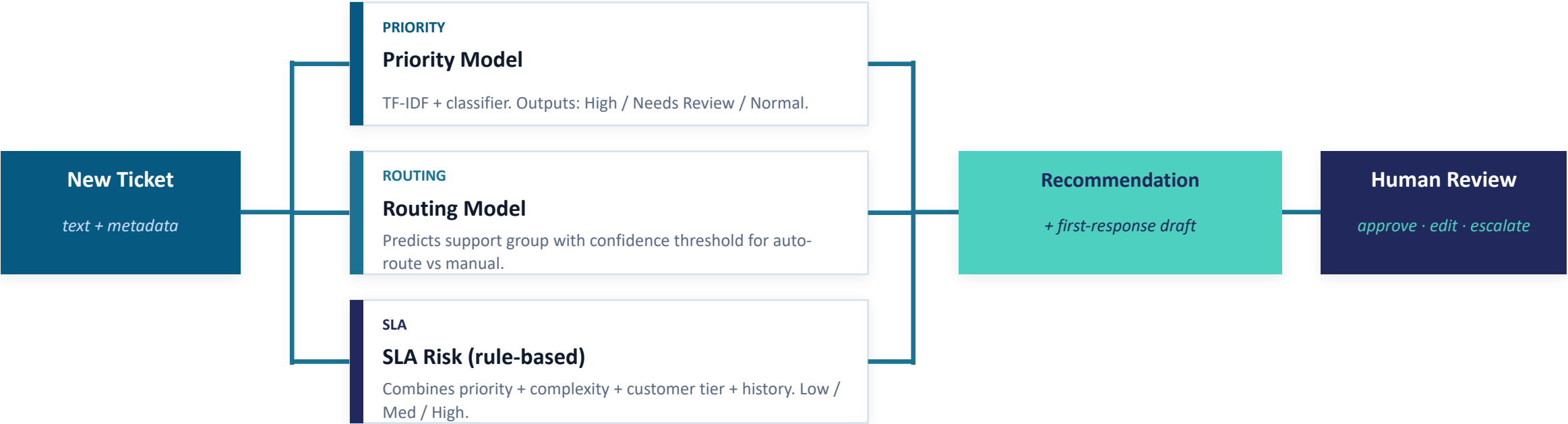
KEY DESIGN DECISION

ML had no real signal. Replaced with a transparent rule-based score over priority, complexity, customer tier, and history.

All three models combine into one recommendation engine that outputs priority, routing, SLA risk, and a first-response draft per ticket.

One ticket's journey through TriageIQ.

A new ticket flows through three AI components in parallel, then converges on a single recommendation reviewed by a human agent.



Parallel inference

all three components score the ticket simultaneously

Confidence thresholds

high-confidence outputs auto-applied; low-confidence routed to review

Human-in-the-loop

no action is taken without an agent confirming

RESULTS

Model performance — and the deliberate tradeoffs.

Tested on held-out outer-test data. Numbers come directly from our notebook outputs.

PRIORITY MODEL · SGD Logistic + TF-IDF

95%

High-priority recall

catches 95 of every 100 urgent tickets

Overall accuracy	55%
High precision	46%
High F1	0.62
Not-High precision / recall	90% / 29%

Recall-first by design: missing an urgent ticket costs more than over-flagging.

ROUTING MODEL · Grouped SGD Logistic + TF-IDF

81%

Accuracy

0.80

Weighted F1

F1 by routing group

Confidence thresholds: high-confidence routes auto-applied, low-confidence sent to manual review.



SLA prediction was the honest negative result.

Logistic Regression and XGBoost both scored ROC-AUC $\approx$ 0.50 (random). We replaced ML with a transparent rule-based SLA risk score using priority, complexity, customer tier, and history.

Where the model is fair — and where it isn't.



Language bias

German tickets are misrouted 5–31% more often than English, especially for business support. The TF-IDF model embeds DE-specific terms into wrong routes. Mitigation: language-agnostic training data.

Priority Model — TPR vs FPR by language

True positive rate (catching High) and false positive rate, broken out by ticket language.



Queue imbalance

IT / Product / Technical Support route correctly >95% of the time, but Customer Service tickets route correctly only 24.6%, and HR just 17.4% — these queues are too small in the training data.



Generalization gap

On the 200k dataset, every category predicts "high priority" at ~40% — a coin-flip. The model overfits the smaller aa_dataset and does not transfer cleanly to new distributions.

Takeaway: All categories remain within acceptable accuracy ranges, but generalization beyond the training distribution is the open problem. Reporting these gaps honestly is part of responsible deployment.

What TriageIQ saves a 500-ticket-per-month team.

All figures derived from our team's industry research and reported by published case studies. Conservative midpoint estimates shown.

BEFORE · Manual triage		AFTER · TriageIQ-assisted	
Time per ticket	15–30 min	Time per ticket	5–10 min
Hours per month (500 tickets)	125–150 hrs	Hours per month (500 tickets)	40–55 hrs
Annual triage labor cost	\$300K+	Annual triage labor cost	~\$110K
First-response consistency	Variable	First-response consistency	Uniform

60–70%

reduction in first-response time

~\$190K

annual savings per 500-ticket team

~70 hrs

agent time reclaimed per month

Savings scale with ticket volume — larger enterprises see proportionally larger gains. Estimates exclude integration costs and SLA-breach risk reduction.

TAKEAWAY

TriageIQ shows where AI helps — and where it shouldn't.



What worked

- Priority detection with TF-IDF + recall-focused thresholds (95% High recall)
- Queue routing after grouping overlapping labels (80.6% accuracy, 95.6% at high confidence)
- Human-in-the-loop design that reviews uncertain predictions instead of automating them away



What didn't and why that matters

- SLA breach prediction (ROC-AUC ≈ 0.5) — replaced with a transparent rule-based score
- Cross-dataset generalization (200k dataset predicts ~40% "high" for every category)
- Routing of low-volume queues (Customer Service, HR) due to training data imbalance